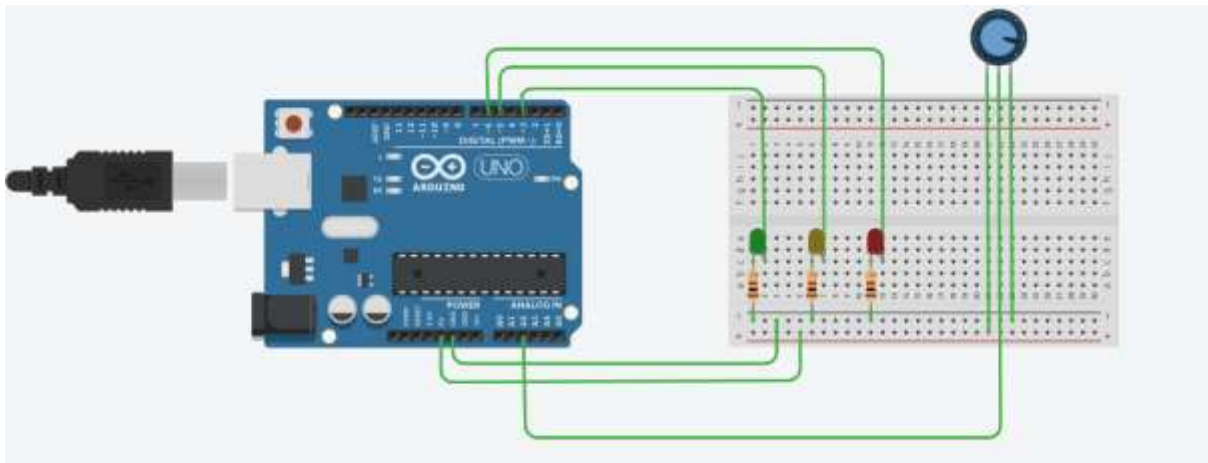




```
int potPin = A2;
int greenLED = 3;
int yellowLED = 5;
int redLED = 6;

void setup() {
  pinMode(greenLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(redLED, OUTPUT);
  Serial.begin(9600);
}

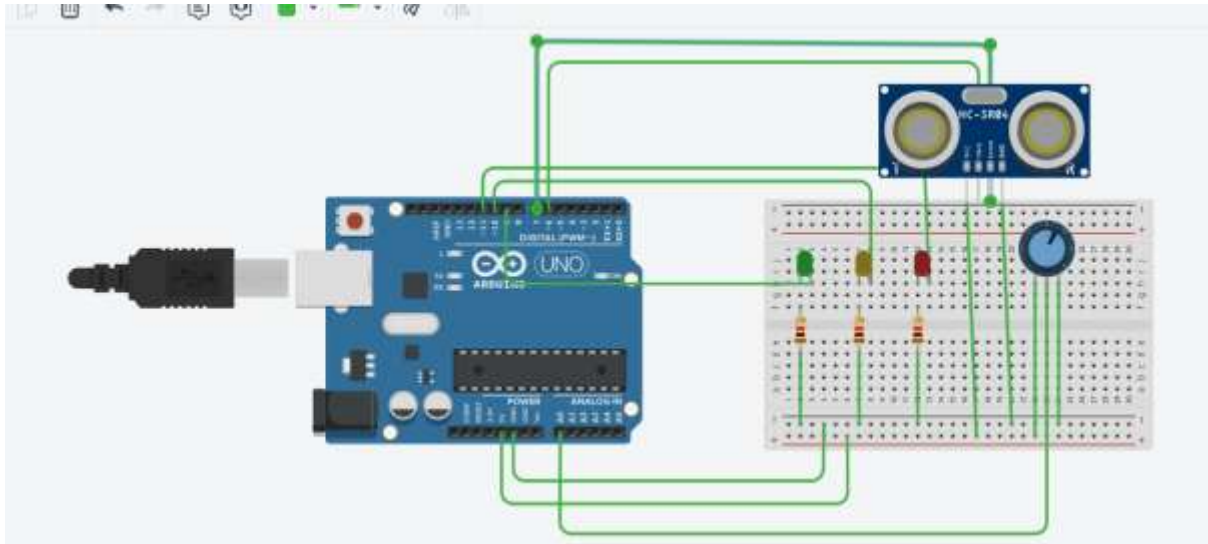
void loop() {
  int potValue = analogRead(potPin);
  if (potValue >= 0 && potValue <= 340) {
    int brightness = map(potValue, 0, 340, 0, 255);
    analogWrite(greenLED, brightness);
    analogWrite(yellowLED, 0);
    analogWrite(redLED, 0);
    Serial.print("Potentiometer: ");
    Serial.print(potValue);
```

```
Serial.print(" | Green LED ON | Brightness: ");  
Serial.println(brightness);  
}
```

```
else if (potValue >= 341 && potValue <= 680) {  
    int brightness = map(potValue, 341, 680, 0, 255);  
    analogWrite(greenLED, 0);  
    analogWrite(yellowLED, brightness);  
    analogWrite(redLED, 0);  
    Serial.print("Potentiometer: ");  
    Serial.print(potValue);  
    Serial.print(" | Yellow LED ON | Brightness: ");  
    Serial.println(brightness);  
}
```

```
else {  
    int brightness = map(potValue, 681, 1023, 0, 255);  
    analogWrite(greenLED, 0);  
    analogWrite(yellowLED, 0);  
    analogWrite(redLED, brightness);  
    Serial.print("Potentiometer: ");  
    Serial.print(potValue);  
    Serial.print(" | Red LED ON | Brightness: ");  
    Serial.println(brightness);  
}
```

```
delay(100);  
}
```



```
int trigPin = 6;

int echoPin = 7;

int potPin = A0;

int greenLED = 9;

int yellowLED = 10;

int redLED = 11;

void setup() {

  pinMode(trigPin, OUTPUT);

  pinMode(echoPin, INPUT);

  pinMode(greenLED, OUTPUT);

  pinMode(yellowLED, OUTPUT);

  pinMode(redLED, OUTPUT);

  Serial.begin(9600);

}

void loop() {

  int potValue = analogRead(potPin);

  int warningDistance = map(potValue, 0, 1023, 10, 50);

  digitalWrite(trigPin, LOW);

  delayMicroseconds(2);
```

```
digitalWrite(trigPin, HIGH);  
delayMicroseconds(10);  
digitalWrite(trigPin, LOW);  
long duration = pulseIn(echoPin, HIGH);  
float distance = duration * 0.0343 / 2;  
if (distance > warningDistance) {  
    digitalWrite(greenLED, HIGH);  
    digitalWrite(yellowLED, LOW);  
    digitalWrite(redLED, LOW);  
    Serial.println("State: SAFE - Green LED ON");  
}  
else if (distance > warningDistance / 2.0) {  
    digitalWrite(greenLED, LOW);  
    digitalWrite(yellowLED, HIGH);  
    digitalWrite(redLED, LOW);  
    Serial.println("State: WARNING - Yellow LED ON");  
}  
else {  
    digitalWrite(greenLED, LOW);  
    digitalWrite(yellowLED, LOW);  
    digitalWrite(redLED, HIGH);  
    Serial.println("State: DANGER - Red LED ON");  
}  
Serial.print("Potentiometer: ");  
Serial.print(potValue);  
Serial.print(" | Warning Distance: ");  
Serial.print(warningDistance);  
Serial.print(" cm");
```

```
Serial.print(" | Actual Distance: ");  
Serial.print(distance);  
Serial.println(" cm");  
delay(500);  
}
```

3.

```
float mapFloat(float x, float in_min, float in_max,  
               float out_min, float out_max)  
{  
    return (x - in_min) * (out_max - out_min) /  
           (in_max - in_min) + out_min;  
}  
  
void setup()  
{  
    Serial.begin(9600);  
}  
  
void loop()  
{  
    int sensorValue = analogRead(A0);  
    float mappedValue = mapFloat(sensorValue, 0, 1023, 0.0, 5.0);  
    Serial.print("Sensor Value: ");  
    Serial.print(sensorValue);  
    Serial.print(" Mapped Value: ");  
    Serial.println(mappedValue);  
    delay(500);  
}
```